

Data Assessment – Guardian Mutual Assurance (GMA)

This notebook performs data exploration and assessment for the Claims, Policyholders, and Third-Parties datasets.

Objectives:

- Understand data structure and types
- Identify missing values
- Explore target variable distribution
- Detect extreme outliers
- Map dataset relationships
- Highlight key data issues

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load datasets
claims = pd.read_excel("claims.xlsx")
policyholders = pd.read_excel("policyholders.xlsx")
third_parties = pd.read_excel("third_parties.xlsx")
```

Initial Exploration

We first inspect the data using `.head()`, `.info()`, and `.describe()` to understand:

- Column names
- Data types
- Summary statistics

```
# Claims
print("Claims Table Head:")
display(claims.head())
print("Claims Info:")
display(claims.info())
print("Claims Describe:")
display(claims.describe())

# Policyholders
print("Policyholders Table Head:")
display(policyholders.head())
print("Policyholders Info:")
display(policyholders.info())
print("Policyholders Describe:")
display(policyholders.describe())

# Third-Parties
print("Third-Parties Table Head:")
display(third_parties.head())
print("Third-Parties Info:")
display(third_parties.info())
print("Third-Parties Describe:")
display(third_parties.describe())
```

Claims Table Head:

```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	Claim_ID	Policy_ID	Accident_Date	FNOL_Date	Claim_Type	Claim_Complexity	Fraud_Flag	Litigation_Flag	Estimated_Claim_Amount	...
--	----------	-----------	---------------	-----------	------------	------------------	------------	-----------------	------------------------	-----

	Claim_ID	Policy_ID	Accident_Date	FNOL_Date	Claim_Type	Claim_Complexity	Fraud_Flag	Litigation_Flag	Estimated_Claim_Amount	
0	CLM30000	POL14506	2019-12-19	2019-12-19	Theft	Medium	False	True	5243	2
1	CLM30001	POL14338	2018-12-30	2018-12-31	Collision	Low	False	False	3934	2
2	CLM30002	POL13575	2021-10-19	2021-10-19	Other	Medium	False	False	153631	1
3	CLM30003	POL10138	2021-06-18	2021-06-18	Weather	Low	False	False	2812	1
4	CLM30004	POL12316	2021-03-21	2021-03-24	Theft	Low	False	False	5094	4

Claims Info:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8000 entries, 0 to 7999
Data columns (total 13 columns):
Column Non-Null Count Dtype
--- -----
0 Claim_ID 8000 non-null object
1 Policy_ID 8000 non-null object
2 Accident_Date 8000 non-null datetime64[ns]
3 FNOL_Date 8000 non-null datetime64[ns]
4 Claim_Type 8000 non-null object
5 Claim_Complexity 8000 non-null object
6 Fraud_Flag 8000 non-null bool
7 Litigation_Flag 8000 non-null bool
8 Estimated_Claim_Amount 8000 non-null int64
9 Ultimate_Claim_Amount 7575 non-null float64
10 Severity_Band 8000 non-null object
11 Settlement_Date 7575 non-null datetime64[ns]
12 Status 8000 non-null object
dtypes: bool(2), datetime64[ns](3), float64(1), int64(1), object(6)
memory usage: 703.3+ KB

None

Claims Describe:

```
.dataframe tbody tr th {  
    vertical-align: top;  
}  
  
.dataframe thead th {  
    text-align: right;  
}
```

	Accident_Date	FNOL_Date	Estimated_Claim_Amount	Ultimate_Claim_Amount	Settlement_Date
count	8000	8000	8.000000e+03	7.575000e+03	7575
mean	2022-04-02 01:52:51.600000	2022-04-03 20:20:45.600000256	1.447988e+04	1.311868e+04	2022-07-15 14:21:54.534653440
min	2018-09-19 00:00:00	2018-09-19 00:00:00	5.500000e+02	3.320000e+02	2018-10-25 00:00:00
25%	2020-06-20 00:00:00	2020-06-22 18:00:00	2.205750e+03	1.578000e+03	2020-10-11 12:00:00
50%	2022-04-06 00:00:00	2022-04-07 00:00:00	4.416000e+03	3.409000e+03	2022-07-14 00:00:00
75%	2024-01-05 00:00:00	2024-01-06 12:00:00	1.167475e+04	9.744000e+03	2024-04-17 00:00:00
max	2025-09-17 00:00:00	2025-09-22 00:00:00	1.064239e+06	1.005590e+06	2027-03-29 00:00:00
std	NaN	NaN	3.873981e+04	3.810405e+04	NaN

Policyholders Table Head:

```
.dataframe tbody tr th {  
    vertical-align: top;  
}
```

```
.dataframe thead th {
  text-align: right;
}
```

	Policy_ID	Customer_ID	Age_of_Driver	Gender	Occupation	Region	Annual_Mileage	Driving_Experience_Years	Vehicle_Type	Vehic
0	POL10000	CUST20000	56	Female	Retired	Newcastle	7552	36	Sedan	10
1	POL10001	CUST20001	53	Female	Unemployed	Bristol	13275	31	Motorcycle	11
2	POL10002	CUST20002	19	Female	Unemployed	London	12967	0	Sedan	9
3	POL10003	CUST20003	77	Female	Retired	Birmingham	4346	56	Hatchback	2
4	POL10004	CUST20004	24	Male	Employed	Manchester	9598	2	Sedan	14

```
Policyholders Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Policy_ID              5000 non-null   object
1   Customer_ID            5000 non-null   object
2   Age_of_Driver          5000 non-null   int64
3   Gender                 5000 non-null   object
4   Occupation              5000 non-null   object
5   Region                 5000 non-null   object
6   Annual_Mileage          5000 non-null   int64
7   Driving_Experience_Years 5000 non-null   int64
8   Vehicle_Type           5000 non-null   object
9   Vehicle_Age            5000 non-null   int64
10  Credit_Score_Band       5000 non-null   object
dtypes: int64(4), object(7)
memory usage: 429.8+ KB

None

Policyholders Describe:
```

```
.dataframe tbody tr th {
  vertical-align: top;
}

.dataframe thead th {
  text-align: right;
}
```

	Age_of_Driver	Annual_Mileage	Driving_Experience_Years	Vehicle_Age
count	5000.0000	5000.000000	5000.000000	5000.0000
mean	48.7524	12125.190800	28.817600	9.9910
std	17.9704	4036.274571	17.919335	6.0998
min	18.0000	500.000000	0.000000	0.0000
25%	33.0000	9388.250000	13.000000	5.0000
50%	49.0000	12082.000000	29.000000	10.0000
75%	65.0000	14860.750000	45.000000	15.0000
max	79.0000	27410.000000	61.000000	20.0000

```
Third-Parties Table Head:
```

```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	Claim_ID	TP_ID	ThirdParty_Role	TP_Injury_Severity
0	CLM30000	TP40000	Pedestrian	Minor
1	CLM30002	TP40001	Passenger	Minor
2	CLM30007	TP40002	Pedestrian	Minor
3	CLM30012	TP40003	Pedestrian	Minor
4	CLM30015	TP40004	Driver	Minor

```
Third-Parties Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2410 entries, 0 to 2409
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Claim_ID              2410 non-null   object
1   TP_ID                 2410 non-null   object
2   ThirdParty_Role        2410 non-null   object
3   TP_Injury_Severity     2410 non-null   object
dtypes: object(4)
memory usage: 75.4+ KB

None

Third-Parties Describe:
```

```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	Claim_ID	TP_ID	ThirdParty_Role	TP_Injury_Severity
count	2410	2410	2410	2410
unique	1997	2410	3	3
top	CLM37990	TP40000	Passenger	Minor
freq	2	1	836	2050

Missing Values Analysis

We examine missing values per column to determine completeness and potential issues.

```
def missing_values(df):
    return (
        df.isnull()
        .sum()
        .to_frame("Missing_Count")
        .assign(Missing_Percent=lambda x: x.Missing_Count / len(df) * 100)
        .sort_values("Missing_Percent", ascending=False)
    )

# Apply function
```

```
missing_claims = missing_values(claims)
missing_policyholders = missing_values(policyholders)
missing_third_parties = missing_values(third_parties)

display(missing_claims)
display(missing_policyholders)
display(missing_third_parties)
```

```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	Missing_Count	Missing_Percent
Ultimate_Claim_Amount	425	5.3125
Settlement_Date	425	5.3125
Claim_ID	0	0.0000
Policy_ID	0	0.0000
Accident_Date	0	0.0000
FNOL_Date	0	0.0000
Claim_Type	0	0.0000
Claim_Complexity	0	0.0000
Fraud_Flag	0	0.0000
Litigation_Flag	0	0.0000
Estimated_Claim_Amount	0	0.0000
Severity_Band	0	0.0000
Status	0	0.0000

```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	Missing_Count	Missing_Percent
Policy_ID	0	0.0
Customer_ID	0	0.0
Age_of_Driver	0	0.0
Gender	0	0.0
Occupation	0	0.0
Region	0	0.0
Annual_Mileage	0	0.0
Driving_Experience_Years	0	0.0
Vehicle_Type	0	0.0
Vehicle_Age	0	0.0
Credit_Score_Band	0	0.0

```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	Missing_Count	Missing_Percent
Claim_ID	0	0.0
TP_ID	0	0.0
ThirdParty_Role	0	0.0
TP_Injury_Severity	0	0.0

Target Variable Distribution

- The `Ultimate_Claim_Amount` is highly right-skewed.
- Most claims are low-cost, but a few very high-cost claims exist.
- These outliers are valid and retained.

```
plt.figure(figsize=(12, 5))
plt.hist(claims["Ultimate_Claim_Amount"].dropna()/1000, bins=50, log=True, color='skyblue', edgecolor='black')
plt.xlabel("Ultimate Claim Amount (£k)")
plt.ylabel("Log Frequency")
plt.title("Ultimate Claim Amount Distribution (Log Scale) - To Detect Outliers")
plt.show()
```



Schema Mapping

- `Claim_ID` links Claims → Third-Parties (one-to-many)
- `Policy_ID` links Claims → Policyholders (one-to-one)
- `Accident_Date` links Claims → Weather data (optional join)

Key Data Issues

1. Missing `Ultimate_Claim_Amount` (~5.31%) – removed from training dataset.
2. Missing `Settlement_Date` (~5.31%) – retained as NaT to represent open claims.
3. Highly skewed target variable – requires robust modeling.
4. FNOL-stage data uncertainty – incomplete initial reports.

02_data_assessment

1. Missing Values Assessment

An analysis of missing values was conducted across all datasets.

Missing Values Summary Dataset Column Missing Count Missing % Treatment Claims Ultimate_Claim_Amount 425 5.31% Rows excluded from model training Claims Settlement_Date 425 5.31% Retained as NaT (open claims) Policyholders – 0 0% No action required Third-Parties – 0 0% No action required

Key Observations: No missing values were found in the Policyholders or Third-Parties datasets. Missing values in the Claims dataset are limited to settlement-related fields. The missingness is expected and reflects claims that are still open.

2. Target Variable Distribution & Skewness

The distribution of the `Ultimate_Claim_Amount` was examined using histograms with log-scaled frequency.

Observations: The target variable exhibits a strong right-skewed distribution. Most claims are low-cost and occur frequently. A small number of high-cost claims create a long right tail. Implications: Extreme values (outliers) are expected in insurance claim data. Models should be robust to skewness and outliers. Consideration may be given to transformations (e.g., log transformation) during modeling.

3. Presence of Extreme Outliers

Visual inspection of the target variable confirmed the presence of extreme high-value claims. Outliers represent rare but financially significant events. These claims are critical for reserve estimation and solvency planning. Outliers were not removed, as they reflect real-world insurance risk.

4. Key Data Issues Identified

Issue Description Impact Incomplete settlement dates Missing Settlement_Date for open claims Expected; handled using NaT Missing target values Ultimate_Claim_Amount missing for unsettled claims Rows excluded from training Highly skewed target Long right tail in claim amounts Requires robust modeling FNOL data uncertainty Limited information available at FNOL stage Motivates predictive modeling

5. Summary

The dataset is largely complete, with missing values confined to settlement-related fields in the Claims dataset. The target variable is highly right-skewed with extreme outliers, which is typical in insurance data. Missing values were handled using domain-appropriate strategies to ensure modeling integrity, regulatory compliance, and realistic FNOL-stage prediction.

